

R Final Project

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#Library Installation

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.1      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.2.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

#Data Loading ##Loading each monthly sales

```
jan_sales <- read_csv("data/raw/2019-sales-monthly/201901-sales.csv")
```

```
## Rows: 9723 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(jan_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>          <chr>             <chr>        <chr>
## 1 141234     iPhone           1                700          01/22/19 21~
## 2 141235     Lightning Charging Ca~ 1                14.95         01/28/19 14~
## 3 141236     Wired Headphones    2                11.99         01/17/19 13~
## 4 141237     27in FHD Monitor    1                149.99        01/05/19 20~
## 5 141238     Wired Headphones    1                11.99         01/25/19 11~
## 6 141239     AAA Batteries (4-pack) 1                2.99          01/29/19 20~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
feb_sales <- read_csv("data/raw/2019-sales-monthly/201902-sales.csv")
```

```
## Rows: 12036 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(feb_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>          <chr>            <chr>        <chr>
## 1 150502     iPhone             1              700          02/18/19 01~
## 2 150503     AA Batteries (4-pack) 1              3.84          02/13/19 07~
## 3 150504     27in 4K Gaming Monitor 1             389.99          02/18/19 09~
## 4 150505     Lightning Charging Ca~ 1             14.95          02/02/19 16~
## 5 150506     AA Batteries (4-pack) 2              3.84          02/28/19 20~
## 6 150507     Lightning Charging Ca~ 1             14.95          02/24/19 18~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
mar_sales <- read_csv("data/raw/2019-sales-monthly/201903-sales.csv")
```

```
## Rows: 15226 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(mar_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>          <chr>            <chr>        <chr>
## 1 162009     iPhone             1              700          03/28/19 20~
## 2 162009     Lightning Charging Ca~ 1             14.95          03/28/19 20~
## 3 162009     Wired Headphones      2             11.99          03/28/19 20~
## 4 162010     Bose SoundSport Headp~ 1             99.99          03/17/19 05~
## 5 162011     34in Ultrawide Monitor 1            379.99          03/10/19 00~
## 6 162012     AA Batteries (4-pack) 1              3.84          03/20/19 21~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
apr_sales <- read_csv("data/raw/2019-sales-monthly/201904-sales.csv")
```

```
## Rows: 18383 Columns: 6
## -- Column specification -----
```

```
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(apr_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>             <chr>      <chr>
## 1 176558    USB-C Charging Cable 2             11.95      04/19/19 08~
## 2 <NA>      <NA>      <NA>             <NA>      <NA>
## 3 176559    Bose SoundSport Headp~ 1             99.99      04/07/19 22~
## 4 176560    Google Phone          1             600        04/12/19 14~
## 5 176560    Wired Headphones      1             11.99      04/12/19 14~
## 6 176561    Wired Headphones      1             11.99      04/30/19 09~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
may_sales <- read_csv("data/raw/2019-sales-monthly/201905-sales.csv")
```

```
## Rows: 16635 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(may_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>             <chr>      <chr>
## 1 194095    Wired Headphones      1             11.99      05/16/19 17~
## 2 194096    AA Batteries (4-pack) 1             3.84       05/19/19 14~
## 3 194097    27in FHD Monitor      1            149.99     05/24/19 11~
## 4 194098    Wired Headphones      1             11.99     05/02/19 20~
## 5 194099    AAA Batteries (4-pack) 2             2.99      05/11/19 22~
## 6 194100    iPhone                1            700.0      05/10/19 19~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
jun_sales <- read_csv("data/raw/2019-sales-monthly/201906-sales.csv")
```

```
## Rows: 13622 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(jun_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>             <chr>       <chr>
## 1 209921    USB-C Charging Cable 1             11.95       06/23/19 19~
## 2 209922    Macbook Pro Laptop   1             1700.0      06/30/19 10~
## 3 209923    ThinkPad Laptop      1             999.99      06/24/19 20~
## 4 209924    27in FHD Monitor     1             149.99      06/05/19 10~
## 5 209925    Bose SoundSport Headp~ 1             99.99       06/25/19 18~
## 6 209926    Apple AirPods Headpho~ 1             150.0       06/28/19 20~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
jul_sales <- read_csv("data/raw/2019-sales-monthly/201907-sales.csv")
```

```
## Rows: 14371 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(jul_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>             <chr>       <chr>
## 1 222910    Apple AirPods Headpho~ 1             150         07/26/19 16~
## 2 222911    Flatscreen TV          1             300         07/05/19 08~
## 3 222912    AA Batteries (4-pack) 1             3.84        07/29/19 12~
## 4 222913    AA Batteries (4-pack) 1             3.84        07/28/19 10~
## 5 222914    AAA Batteries (4-pack) 5             2.99        07/31/19 02~
## 6 222915    Bose SoundSport Headp~ 1             99.99       07/03/19 18~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
aug_sales <- read_csv("data/raw/2019-sales-monthly/201908-sales.csv")
```

```
## Rows: 12011 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(aug_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
```

```
##   <chr>      <chr>      <chr>      <chr>      <chr>
## 1 236670    Wired Headphones 2      11.99    08/31/19 22~
## 2 236671    Bose SoundSport Headp~ 1      99.99    08/15/19 15~
## 3 236672    iPhone 1      700.0    08/06/19 14~
## 4 236673    AA Batteries (4-pack) 2      3.84     08/29/19 20~
## 5 236674    AA Batteries (4-pack) 2      3.84     08/15/19 19~
## 6 236675    Wired Headphones 1      11.99    08/02/19 23~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
sep_sales <- read_csv("data/raw/2019-sales-monthly/201909-sales.csv")
```

```
## Rows: 11686 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(sep_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>      <chr>      <chr>
## 1 248151    AA Batteries (4-pack) 4      3.84     09/17/19 14~
## 2 248152    USB-C Charging Cable 2      11.95    09/29/19 10~
## 3 248153    USB-C Charging Cable 1      11.95    09/16/19 17~
## 4 248154    27in FHD Monitor 1      149.99   09/27/19 07~
## 5 248155    USB-C Charging Cable 1      11.95    09/01/19 19~
## 6 248156    34in Ultrawide Monitor 1      379.99   09/13/19 14~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
oct_sales <- read_csv("data/raw/2019-sales-monthly/201910-sales.csv")
```

```
## Rows: 20379 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(oct_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>      <chr>      <chr>
## 1 259358    34in Ultrawide Monitor 1      379.99   10/28/19 10~
## 2 259359    27in 4K Gaming Monitor 1      389.99   10/28/19 17~
## 3 259360    AAA Batteries (4-pack) 2      2.99     10/24/19 17~
## 4 259361    27in FHD Monitor 1      149.99   10/14/19 22~
```

```
## 5 259362      Wired Headphones      1      11.99      10/07/19 16~
## 6 259363      AAA Batteries (4-pack) 1      2.99      10/01/19 18~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
nov_sales <- read_csv("data/raw/2019-sales-monthly/201911-sales.csv")
```

```
## Rows: 17661 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(nov_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>      <chr>      <chr>
## 1 278797      Wired Headphones      1      11.99      11/21/19 09~
## 2 278798      USB-C Charging Cable  2      11.95      11/17/19 10~
## 3 278799      Apple AirPods Headpho~ 1      150.0      11/19/19 14~
## 4 278800      27in FHD Monitor      1      149.99      11/25/19 22~
## 5 278801      Bose SoundSport Headp~ 1      99.99      11/09/19 13~
## 6 278802      USB-C Charging Cable  1      11.95      11/14/19 20~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
dec_sales <- read_csv("data/raw/2019-sales-monthly/201912-sales.csv")
```

```
## Rows: 25117 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(dec_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>      <chr>      <chr>
## 1 295665      Macbook Pro Laptop      1      1700      12/30/19 00::~
## 2 295666      LG Washing Machine      1      600.0      12/29/19 07::~
## 3 295667      USB-C Charging Cable  1      11.95      12/12/19 18::~
## 4 295668      27in FHD Monitor      1      149.99      12/22/19 15::~
## 5 295669      USB-C Charging Cable  1      11.95      12/18/19 12::~
## 6 295670      AA Batteries (4-pack) 1      3.84      12/31/19 22::~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
## Combine the 12 months data using bind_rows
```

```
all_sales_2019_combine <- bind_rows(jan_sales, feb_sales, mar_sales,
                                     apr_sales, may_sales, jun_sales,
                                     jul_sales, aug_sales, sep_sales,
                                     oct_sales, nov_sales, dec_sales)

head(all_sales_2019_combine)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>             <chr>      <chr>
## 1 141234    iPhone          1                700        01/22/19 21~
## 2 141235    Lightning Charging Ca~ 1                14.95       01/28/19 14~
## 3 141236    Wired Headphones  2                11.99       01/17/19 13~
## 4 141237    27in FHD Monitor  1                149.99      01/05/19 20~
## 5 141238    Wired Headphones  1                11.99       01/25/19 11~
## 6 141239    AAA Batteries (4-pack) 1                2.99        01/29/19 20~
## # i 1 more variable: 'Purchase Address' <chr>
```